RISE RESEARCH INNOVATION SCHOLARSHIP ENTREPRENEURSHIP
Mentor Name. : Eric Gerber
Subject/Category : Machine Learning
Class : DS 4420 — Machine Learning and Data Mining 2
Northeastern University

# DS4420 project — Stock Forecasting with Time Series and Bayesian Linear Regression Model

Jialin Weng, Mengyang Wang

## Introduction

Predicting stock prices is an important topic in finance, but it is difficult because stock prices change a lot over time. In this project, we try to use past closing prices to predict future prices for eight large technology stocks. We use two methods from this course, a SARIMAX time series model and a Bayesian linear regression model, to study this problem. Our goal is to examine how each model performs on the same dataset.

## Motivation and Data

We study large technology companies because they are important in the market and have long historical price records. The data are from Yahoo Finance and include daily closing prices for NVDA, AAPL, MSFT, AVGO, MU, ORCL, AMD, and TSM. The sample period is from 2015 to 2025. We use this dataset to see if these two methods can capture useful patterns in stock prices.

Example source link:
https://finance.yahoo.com/quote/NVDA/history/?period1=1420070400&period2=1767225600

## Method

**SARIMAX Time Series Model**

- Yahoo Finance dataset with 8 stocks
- log-transformed stock prices
- lagged peer-stock prices
- lagged 20-day and 50-day moving averages of the target stock
- one SARIMAX model for each stock
- 80% training and 20% testing
- auto_arima used to choose the order
- RMSE and MAE reported on the original price scale

**Bayesian Linear Regression Model**

- Yahoo Finance dataset with 8 stocks
- log-transformed stock prices
- lagged own-stock price and lagged peer-stock prices
- lagged 20-day and 50-day moving averages of the target stock
- one Bayesian model for each stock
- 80% training and 20% testing
- Gibbs sampling used for model fitting
- posterior predictive mean used for forecasting on the log scale
- forecasts transformed back to the original price scale
- RMSE and MAE reported on the original price scale

## Results

Final Summary Table:

| | Stock | ADF Statistic | ADF p-value | Order | RMSE | MAE |
|---|---|---|---|---|---|---|
| 0 | NVDA | -1.166688 | 0.687759 | (2, 0, 2) | 8.454404 | 6.666853 |
| 1 | AAPL | -0.333347 | 0.920647 | (1, 0, 0) | 8.703246 | 6.840120 |
| 2 | MSFT | -1.009397 | 0.749822 | (2, 0, 1) | 14.765108 | 11.339033 |
| 3 | AVGO | -0.435281 | 0.904039 | (1, 0, 2) | 18.241702 | 12.986977 |
| 4 | MU | -1.070114 | 0.726842 | (3, 0, 2) | 10.542234 | 8.182952 |
| 5 | ORCL | -0.398457 | 0.910356 | (2, 0, 2) | 14.840024 | 10.242821 |
| 6 | AMD | -1.336050 | 0.612522 | (1, 0, 2) | 15.937639 | 12.662384 |
| 7 | TSM | -1.067145 | 0.727995 | (1, 0, 0) | 12.934177 | 9.981193 |

**Figure 1. SARIMAX Results for Eight Technology Stocks**

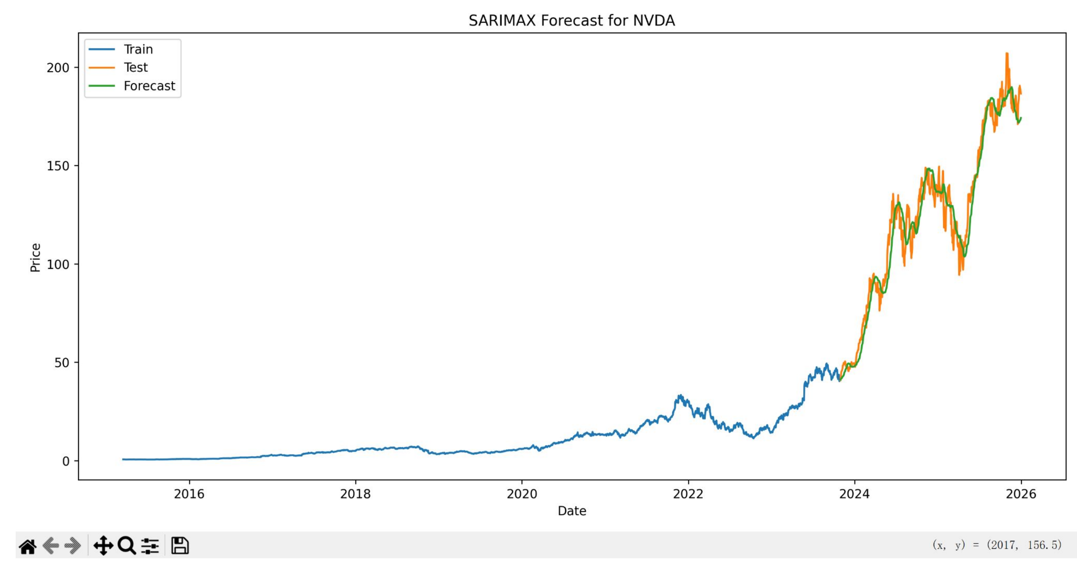


**Figure 2. SARIMAX Forecast for NVDA**

Bayesian Linear Regression Summary:

| | Stock | Posterior mean of Sigma² | RMSE | MAE |
|---|---|---|---|---|
| 1 | AAPL | 0.001264 | 3.642979 | 2.534997 |
| 2 | NVDA | 0.001850 | 3.810938 | 2.770185 |
| 3 | MU | 0.001777 | 4.649755 | 3.132718 |
| 4 | TSM | 0.001304 | 5.251451 | 3.812865 |
| 5 | AMD | 0.002293 | 5.459525 | 3.679547 |
| 6 | MSFT | 0.001231 | 6.027488 | 4.419238 |
| 7 | ORCL | 0.001206 | 6.200605 | 3.298086 |
| 8 | AVGO | 0.001408 | 7.455347 | 4.863672 |

**Figure 3. Bayesian Results for Eight Technology Stocks**

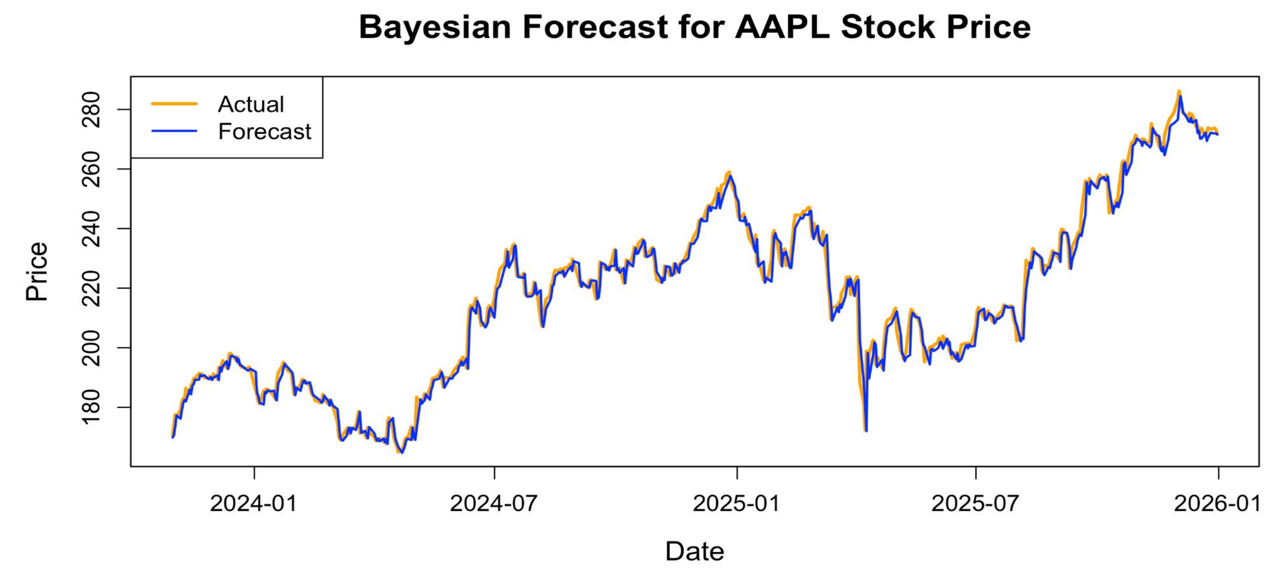


**Figure 4. Bayesian Forecast for AAPL**

The SARIMAX model produced useful forecasts, but performance varied across stocks. NVDA had the lowest forecast error in this model, and AAPL had the second lowest error. The NVDA forecast plot shows that the model followed the main trend of the testing data, although the prediction line was smoother than the actual series.

The Bayesian linear regression model produced useful forecasts overall. Based on the error results, AAPL had the best performance and NVDA had the second lowest error. In the AAPL plot, the forecast was close to the actual series. It followed the general trend well. However, it did not capture some short term fluctuations exactly.

## Discussion and Future Work

Both models made good predictions. SARIMAX uses the stock's own time-series pattern, as well as moving averages and peer stock prices that are lagged. In one regression framework, the Bayesian model uses lagged prices of both its own stock and its peers'. So, SARIMAX might work better for stocks that have a clearer pattern over time, and the Bayesian model might work better for stocks that move more with other stocks.

One problem is that both models only looked at past prices and didn't take into account news, market sentiment, or other outside factors. In subsequent research, we might incorporate stock returns, introduce additional variables, and evaluate these models against more adaptable nonlinear methodologies.

## References

Lestari, I. G. A. N., Deviana, D., & Kusuma, T. M. (2026). A comparative analysis of deep
learning models for stock price prediction. INOVTEK Polbeng – Seri Informatika.
https://jurnal.polbeng.ac.id/index.php/ISI/article/view/1496/657
Reshma, S., Tulasikrishna, G., Prashanth, C. S., Rakesh, C., Venkatesh, K., & Kumar, K. S.
R. (2025). Real-time stock price prediction and market analysis using machine
learning. Proceedings of the International Conference on Research and Development
in Information, Communication, and Computing Technologies.
https://www.scitepress.org/Papers/2025/139421/139421.pdf